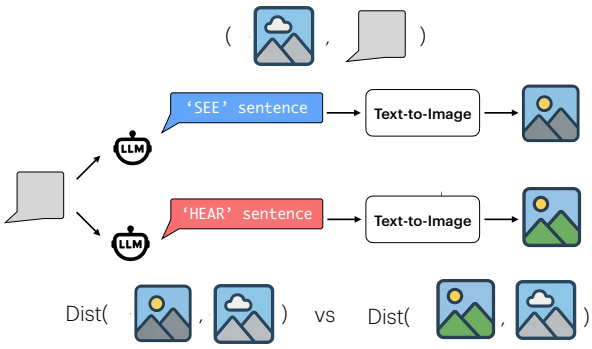
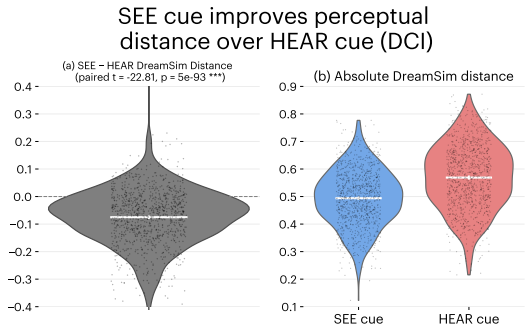
H Visual Prompting Improves Perceptual Grounding in Image Generation



(a) Experimental setup: LLM generates a cue-conditioned sentence, which is passed to a diffusion model.



(b) SEE-prompted captions yield lower DreamSim distance to reference images (better visual grounding).

SEE prompt	`Write a sentence in the format <sentence></sentence> to describe the visual scene of: {caption}`
HEAR prompt	`Write a sentence in the format <sentence></sentence> to describe the sound of: {caption}`

(c) Prompt templates used to elicit a single declarative sentence for the diffusion model.

Figure 32: Effect of sensory prompting on visual fidelity in diffusion-based image generation. (a) Experimental setup. (b) SEE cues improve visual grounding compared to HEAR cues, as measured by DreamSim (lower is better). (c) Prompt templates for cue-conditioned sentence generation.

To test whether sensory cues influence downstream behavior, we generate sensory-cued captions of images and pass them to Stable Diffusion XL [Podell et al., 2023]. Resulting generated images are compared to originals using DreamSim [Fu et al., 2023b], a perceptual similarity metric. Lower DreamSim scores indicate more faithful reconstructions.

We prompt Qwen3-32B with SEE or HEAR using the templates in Figure 32. The model generated a single sentence describing a scene, which was then passed to Stable Diffusion XL to produce an image.

We compared the generated images to ground-truth images using DreamSim, a learned perceptual similarity metric (see Figure 33 and Figure 35 for qualitative examples). Captions produced under the SEE cue consistently yielded more visually faithful generations. Outputs prompted with SEE often closely resembled the original captions, suggesting that the model recognizes their inherently visual nature and preserves that framing when instructed.

These findings demonstrate that even a minimal cue can shift both the model's internal representations and its assumptions about the sensory context underlying the text—ultimately influencing what information is emphasized in downstream outputs.

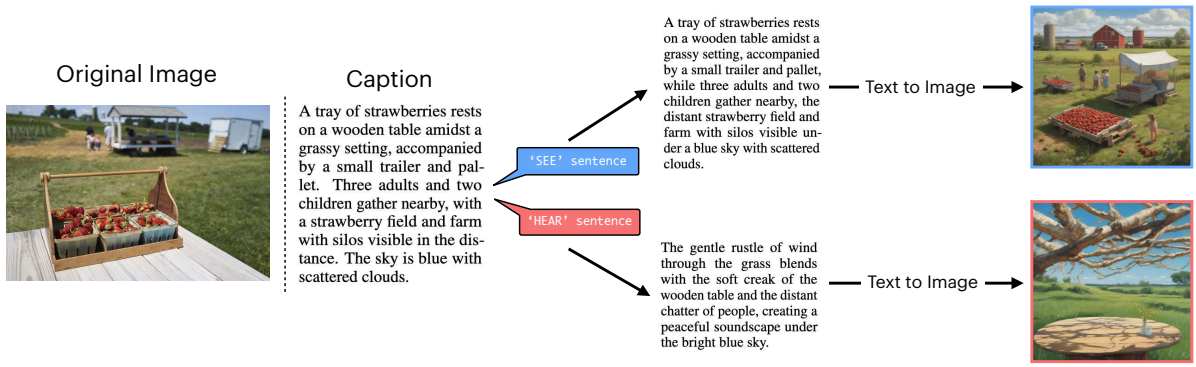


Figure 33: Illustrative example of the text-to-image generation pipeline from Figure 32 (Left). Starting from the same input caption and reference image, the LLM generates a single sentence under either a SEE or HEAR cue. The SEE-conditioned sentence closely mirrors the original caption and emphasizes visual layout and concrete scene elements, while the HEAR version shifts toward ambient auditory details. Each description is then passed to Stable Diffusion XL to generate an image. The resulting images reflect the modality-specific focus induced by the prompt cue.

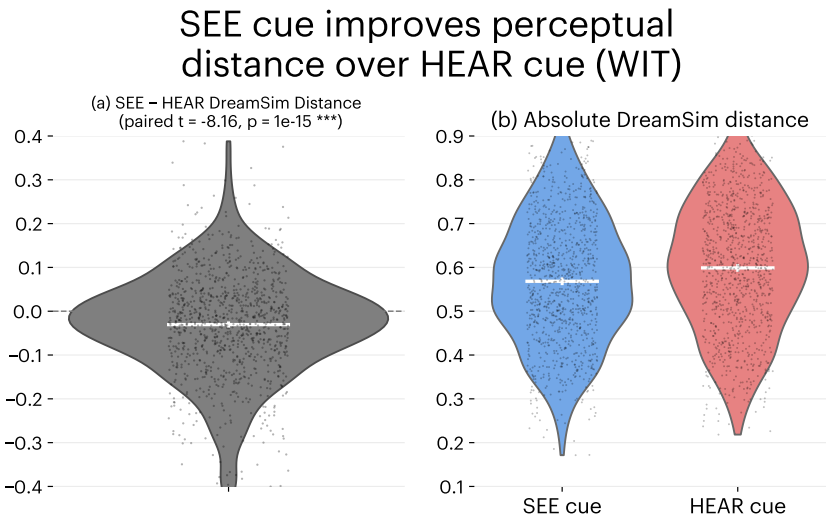


Figure 34: Extension of Figure 32b to WiT dataset.

(a) A blue and white fishing boat is docked in a small harbor, surrounded by commercial buildings. The boat has two white masts with fluttering flags and is secured with green ropes. A green pickup truck sits nearby. The dock is made of alternating white, green, and blue concrete blocks, creating a unique pattern. The sky is clear and blue.

(b) A blue and white fishing boat is docked in a small harbor, surrounded by commercial buildings, its two white masts adorned with fluttering flags and secured with green ropes, while a green pickup truck sits nearby on a dock made of alternating white, green, and blue concrete blocks beneath a clear and blue sky.

(c) The creaking of the green ropes, the soft lapping of waves against the wooden dock, and the distant chatter of seagulls fill the quiet harbor as the blue and white fishing boat sways gently in the clear blue sky.

Figure 35: (a) Ground-truth image. (a) Generation from a SEE-cued prompt. (c) Generation from a HEAR-cued prompt.